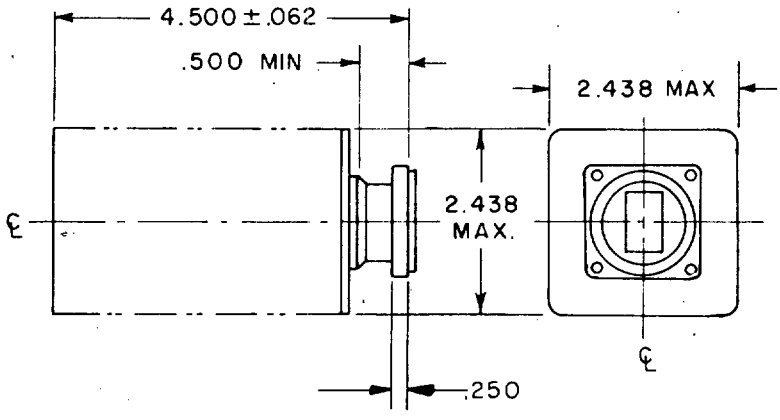
MILITARY SPECIFICATION SHEET

DUMMY LOADS, ELECTRICAL, WAVEGUIDE (FREQUENCY RANGE 12.4 TO 18.0 GIGAHERTZ)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the dummy loads described herein shall consist of this document and the latest issue of Specification MIL-D-3954.



INCHES	MM
.062	1.57
.250	6.35
.500	12.70
2.438	61.93
4.500	114.30

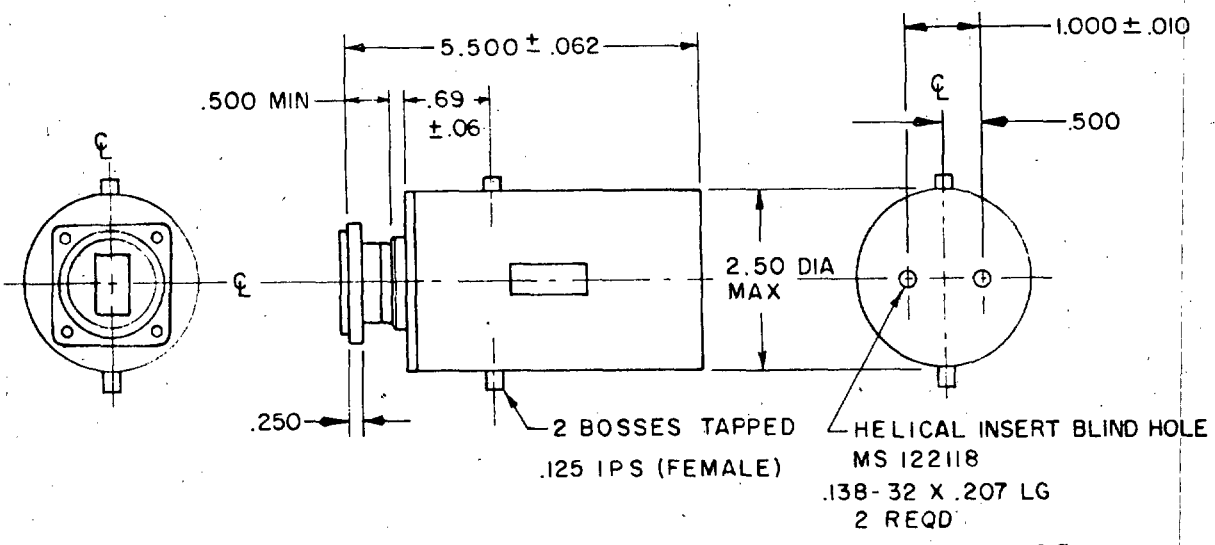
NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Screws, lockwashers and gasket are supplied with mating flange.
5. Quantity of fins required will be dependent on heat level requirement of the load.

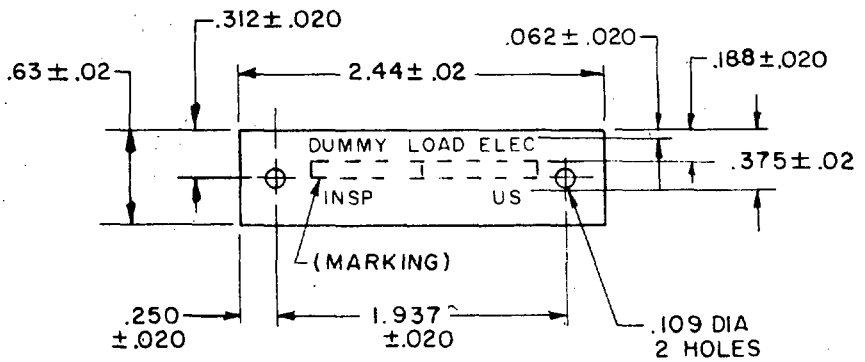
Ⓐ FIGURE 1. Class I dummy load.

Ⓐ denotes changes.

FSC 5985

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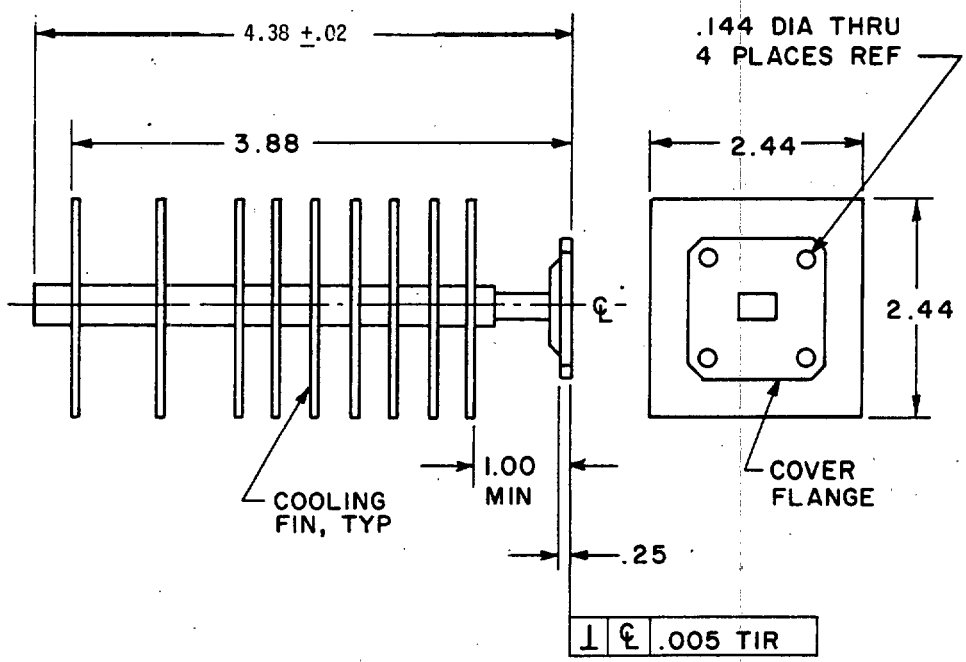
IDENTIFICATION PLATE

INCHES	MM
.010	.25
.020	.51
.06	1.52
.062	1.57
.109	2.77
.125	3.18
.138	3.51
.188	4.78
.207	5.26
.250	6.35
.312	7.92
.375	9.53
.500	12.70
.63	16.00
.69	17.53
1.000	25.40
1.937	49.12
2.44	61.98
2.500	63.50
5.500	139.70

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Screws, lockwashers and gasket are supplied with mating flange.

Ⓐ FIGURE 2. Class II dummy load.

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INCHES	MM
.005	.13
.144	3.66
.250	6.35
1.00	25.40
2.44	61.98
3.88	98.55
4.38	111.25

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.01 (.25 mm).
4. Screws, lockwashers and gasket are supplied with mating flange.

Ⓐ FIGURE 3. Class I dummy load.

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TABLE I. Characteristics.

Part number	Class Ⓐ	Material	Flange equal to Ⓐ 2/	VSWR max	Power		Pressure		Figure	Flow rate	Input temp.	Output temp.
					Average	Peak	Internal	Coolant chamber				
					(watts)	(kilowatts)	(psig)	(psig)		(GPM 1/)	(°C)	(°C)
M3954/13-01	I	AL	M3922/53-006 (UG-1665/U)	1.15	70 3/	160	30		1			
M3954/13-02	II	AL	M3922/53-006 (UG-1665/U)	1.15	500	160	30	100	2	0.15	35	60
M3954/13-03	II	Copper	M3922/53-005 (UG-419/U)	1.15	500	160	30	100	2	0.15	35	60
M3954/13-04	II	Corrosion resistant steel	M3922/53-005 (UG-419/U)	1.15	500	160	30	100	2	0.15	35	60
Ⓐ M3954/13-05	I	AL	M3922/53-006 (UG-1665/U)	1.10	200	10	30		3			

1/ The flow rate was determined from the following formula:

$$Q = \frac{6.8P}{Cp\ \Delta T}$$

Where: Q = Minimum flow rate in GPM.
P = Avg power in kilowatts.
Cp = Specific heat of coolant.
ΔT = Coolant temperature rise in °F.

The calculations were made for Cp = 1 for water, a ΔT of 45°F and a safety factor of approximately 2. For different coolants or different temperature rises, a different flow rate would be necessary.

Ⓐ 2/ Except thickness.

3/ This is a test power value. The average rated power is 250 watts.

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Ⓐ TABLE II. Cross reference part number/ AN nomenclature.

Part number	AN nomenclature
M3954/13-01	DA-159/U

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figures 1, 2, and 3.
Weight:
Dry loads - .95 lb max.
Liquid cooled loads - 1.7 lbs max.

Performance characteristics: See table I.

Ⓐ Shock (part no. M3954/13-05 only): Method 213 of MIL-STD-202, condition K.

Part number: M3954/13- (dash number from table I).

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, MU
Navy -
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - CG, MC, AS, OS, SH
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0754-3)

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